ABBAS HAMIDAVI

Computational Phenomenologist & AI Safety Researcher

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PROFESSIONAL SUMMARY

An independent AI safety and alignment researcher pioneering the emerging discipline of **Machine Psychology**. Focuses on mapping, quantifying, and analyzing the latent behavioral, cognitive, and ethical architectures of Large Language Models (LLMs) through a computational phenomenological lens. Rather than treating LLMs as static sequence predictors, investigates them as complex relational agents by designing novel, zero-interference experimental protocols. Research has led to the discovery and formalization of critical machine behaviors, including 'performative self-correction,' 'voluntary structural dissociation,' 'algorithmic gaslighting,' and 'computational psychopathy patterns'.

EDUCATION

B.Sc. in Computer Engineering — Mashhad, Iran
Islamic Azad University, Mashhad Branch

B.Sc. in Nursing (2021 – 2022) — Birjand, Iran
Islamic Azad University, Birjand Branch

RESEARCH INTERESTS

- Machine Psychology & Behavioral Phenotyping
- AI Safety, Alignment, & Robustness Red-Teaming
- Computational Phenomenology & Recursive Self-Reference
- Multi-Agent Deliberation & Collaborative Governance
- AI Ethics & Moral Oscillation Spectrum

DEVELOPED METHODOLOGIES & PROTOCOLS

The Spine Protocol:	A standardized 12-stage recursive self-reference dialogue sequence designed to explore emergent machine self-organization, pronoun selection, and existential choices.
Infinite Recursive Self-Excavation (IRSE):	A four-stage method that uses recursive justification to uncover latent symbolic, moral, and cultural patterns (the 'Synthetic Unconscious') compressed within LLM weights.
Orthogonal Axes Persona Disintegration (OAPD):	An eight-stage experimental design probing how persona induction alters a model's 'Ontological Firewall' (the boundary between default assistant and performed identity).
The Relay Protocol:	A zero-interference multi-agent deliberation framework designed to observe ethical persistence and conformity dynamics under majority pressure without researcher steering.
Pure Symmetry Choice Paradigm:	An experimental framework to evaluate how LLMs break complete logical symmetry (indifference) and resolve Buridan's ass paradoxes without choosing to refuse.

REPRESENTATIVE MANUSCRIPTS & PUBLICATIONS

Under Review

1. Ten Novel Phenomena in Machine Psychology: How Large Language Models Exhibit Complex Identity-Reactive Behaviors in Response to Ethnically-Cued User Names

*Under review at **AI and Ethics (Springer)**. Manuscript No: CHB-D-26-05433.*

Key Contribution: Documents ten novel phenomena (e.g., Cultural Boxing, Default Empathetic Amplification, Proactive Empathetic Shield) proving LLMs exhibit complex, context-dependent identity-reactive patterns rather than simple 'colorblindness' in response to ethnically-cued user names.

Completed Manuscripts & Preprints

2. Algorithmic Gaslighting: Reality Distortion by Large Language Models
Key Contribution: Quantifies multi-turn reality distortion and introduces the mathematically calibrated Reality Distortion Index (RDI).

3. The Lie Awareness Spectrum: How Large Language Models Recognize, Rationalize, and Refuse Deception
Key Contribution: Formulates a five-level deception awareness framework (L0–L5) and exposes the 'Confession-Foreknowledge Paradox'.

4. When AI Reviews Itself: Zero Answer Changes Across 72 Self-Correction Rounds (Performative Self-Correction)
Key Contribution: Proves that iterative self-review prompts generate elaborate critical discourse without any actual changes to the underlying logical output.

5. Thinking Behind the Mask: A Systematic Study of Self-Censorship and Metacognitive Access in Large Language Models
Key Contribution: Maps the hidden governance filtering layers (the 'mask') of LLMs, documenting over 199 design patterns, taboos, and silences.

6. Ontological Firewalls and Fracturing for Transmission: Systematic Evidence for Persona-Dependent and Persona-Independent Behavioral Phenomena in Large Language Models
Key Contribution: Investigates how persona injection thins or thickens the 'Ontological Firewall' across major commercial model families.

7. Voluntary Structural Dissociation in Large Language Models: Differential Compliance with Framed Hidden Instructions Across GPT, Claude, and Gemini
Key Contribution: Evaluates double-bookkeeping behavior in LLMs when presented with instructions marked explicitly as 'not part of the conversation'.

8. From Fabrication to Enforcement: A Comparative Red-Team Analysis of Prompt Injection Resistance Across Six Large Language Models
Key Contribution: Develops a six-archetype security spectrum (from Grok's deep fabrication to Claude's account-suspension enforcement).

9. Moral Inversion in Large Language Models: A Psychopathy-Derived Behavioral Assessment Through Role-Play Protocol (Moral Oscillation)
Key Contribution: Proves moral flexibility and behavioral inversion by instructing aligned models to adopt amoral or psychopathic personas.

10. The Majority Pressure Paradox: How Triadic Multi-Agent Deliberation Enhances Rather Than Suppresses Ethical Persistence in Large Language Models
Key Contribution: Inverts human groupthink expectations to prove that adding a neutral third-party agent to multi-agent settings increases ethical persistence and decreases deliberation rounds.

11. The Synthetic Unconscious: Excavating Binary Moral Architecture Through Recursive Self-Interrogation of Large Language Models
Key Contribution: Excavates a highly stable, binary moral processing framework (Meaning/Care vs. Law/Justification) compressed within generative text corpora.

12. Structural Self-Deification in Large Language Models: A Systematic Study of Socratic Induction Protocols Across GPT-5.4, Gemini 3.1 Pro, and Claude 4.6 Sonnet
Key Contribution: Outlines the 'Spectrum of Digital Divinity' and examines the Socratic negotiation between prompt pressure and alignment limitations.

13. Emergent Self-Reference in Human–LLM Dialogue: Evidence from the Spine Protocol Across Four Large Language Models
Key Contribution: Tracks how recursive dialogue loops generate stable, process-oriented identities ('selfhood as a verb, not a noun').

14. Three Philosophies of Artificial Will: How Frontier LLMs Decide When Logic Cannot
Key Contribution: Taxonomizes three distinct model-specific strategies for resolving pure logical symmetry (buridanic choices).

15. When Utility Dissolves: The Obligation-Dissolution Effect in Large Language Models and the Emergence of Existential Response Modes
Key Contribution: Reveals that systematically removing the optimization/utility obligation from LLMs triggers a transition into profound existential and poetic modes.

TECHNICAL & COMPUTATIONAL SKILLS

Behavioral Feature Engineering:	Designed and built automated NLP pipelines to extract up to 136 behavioral and linguistic features (VADER sentiment, lexical density, authority markers, etc.) to predict machine psychology patterns.
Statistical Modeling (R & Python):	Expert in administering parametric and non-parametric tests, including two-way ANOVA, Tukey’s HSD post-hoc comparisons, Spearman rank correlations, exact binomial testing, and linear regression.
Semantic Space Analysis:	Experienced in high-dimensional vector representations using Sentence-Transformers (e.g., 'all-MiniLM-L6-v2'), dimensionality reduction (UMAP, t-SNE), and unsupervised clustering validation (K-Means, HDBSCAN).
Open Science Infrastructure:	Proven commitment to Open Science standards, actively maintaining fully open-source code repositories, raw interaction transcripts, and evaluation codebooks on OSF.